25901390 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEMAT201 - TRANSFORMS AND FOURIER SERIES

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type Questions

Answer any **5** questions

Each question carries **2** marks

1. Write the Laplace transform of the unit step function $u(t-a)$. [U] / [CO1]

2. If $\mathcal{L}\{f(t)\} = \dfrac{1}{s^2+1}$, find $\mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\}$. [U] / [CO1]

3. Find the inverse Laplace transform of $\frac{s+1}{s^2}e^{-s}$ using convolution. [A] / [CO2]

4. Find the Laplace transform of the convolution $e^{at} * e^{bt}$. [U] / [CO2]

5. Define Fourier cosine series of perioid $2L$. [K] / [CO3]

6. Define even periodic extension . [K] / [CO3]

7. What is the Fourier transform of the second derivative $f''(x)$. [K] / [CO4]

8. Write the condition for the existence of Fourier cosine transform of a function f . [U] / [CO4]

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **6** marks

9. Find $\mathcal{L}(tsinhat)$ [A] / [CO1]

10. Evaluate $\mathcal{L}^{-1}\left(\frac{4s+1}{s^2-16}\right)$ [U] / [CO1]

11. Find the inverse laplace transform of $\cot^{-1}\left(\frac{s}{\omega}\right)$. [A] / [CO2]

12. Using partial-fraction evaluate the inverse Laplace transform of $\frac{5s}{s(s+3i)(s+4i)}$ [A] / [CO2]

13. Find the Fourier series of the function $f(x) = 7x + 10, -\pi < x < \pi.$ [U] / [CO3]

14. Find the Fourier series expansion of the odd function [U] / [CO3]
$f(x) = 15x, -\pi \leq x \leq \pi.$

15. Find the Fourier Transform of $f(x) = \begin{cases} e^{-kx}, & x > 0 \quad (k > 0) \\ 0, & x < 0 \end{cases}$. [A] / [CO4]

16. Find the Fourier transform of [A] / [CO4]
$f(x) = kx, \quad \text{if } a < x < b \quad \text{and} \quad f(x) = 0, \quad \text{otherwise.}$

[6x5 = 30]

Part C

Essay Type Questions

Answer any **3** questions

Each question carries **10** marks

17. Find the laplace transform of the functions [U] / [CO1]
a) $4e^{5t} + 6t^3 - 3sin4t$ b) $t^4 - sinh2t + 8$

18. Find the inverse Laplace transforms of $\ln\left(\frac{s+1}{s}\right)$ and $\arctan\left(\frac{2}{s}\right)$. [A] / [CO2]

19. Find the inverse Laplace transform of (a) $\frac{(3-2s)}{(s+1)(s+2)(s+6)}$ [A] / [CO2]
(b) $\frac{6s}{(s+1)^2(s-2)^2}$

20. Find the Fourier series of the function [U] / [CO3]
$f(x) = \begin{cases} -1, & -1 < x < 0, \\ 1, & 0 < x < 1, \end{cases}$
with period $p = 2L = 2.$

21. Find the Fourier cosine and Fourier sine transforms of [A] / [CO4]
$f(x) = k$, $0 < x < a$ and $f(x) = 0$, $x > a$.

22. What is convolution? Derive the equation for the Fourier transform of the convolution of two functions $f(x)$ and $g(x)$. [U] / [CO4]

[10x3 = 30]